

SYED ABDULLAH BIN MASOOD

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AI/ML Developer

Karachi, Pakistan

PROFESSIONAL SUMMARY

Results-driven Artificial Intelligence undergraduate and AI/ML Developer with hands-on experience in machine learning, deep learning, MLOps, data engineering, computer vision, multimodal AI, ETL pipelines, developer tooling, and production-oriented AI systems.

Experienced in developing end-to-end machine learning applications, local LLM agents, multimodal deepfake detection systems, automated data pipelines, REST APIs, and deployment-ready ML architectures.

Strong interest in AI safety, multimodal learning, deepfake detection, MLOps, computer vision, LLM-based systems, data engineering, and production-grade machine learning systems.

EDUCATION

Bachelor of Science in Artificial Intelligence

In Progress

Iqra University, Karachi, Pakistan

Relevant Areas: Artificial Intelligence, Machine Learning, Deep Learning, Data Engineering, Databases, Software Engineering, Computer Vision, and Natural Language Processing.

PROFESSIONAL EXPERIENCE

AI/ML & MLOps Internship

10Pearls

- Worked on the development of an end-to-end **Air Quality Index (AQI) Forecasting MLOps system** designed to transform environmental data into production-ready machine learning predictions.
- Developed a modular machine learning architecture covering data ingestion, validation, preprocessing, prediction, API serving, monitoring, and dashboard visualization.
- Built a **FastAPI-based backend** for exposing machine learning predictions through structured REST API endpoints.
- Developed an interactive **Streamlit dashboard** for AQI monitoring, prediction visualization, and model output presentation.
- Implemented reliable API communication mechanisms including asynchronous requests, retry handling, request validation, caching, and failure-management patterns.
- Applied MLOps and software engineering practices including modular project architecture, configuration management, logging, testing, containerization, metrics, observability, and deployment-oriented development.
- Worked with technologies including **Python, FastAPI, Streamlit, Docker, Pydantic, HTTPX, Prometheus, OpenTelemetry, and machine learning frameworks.**

Junior Software Engineer

2 Months

IoT Systems

- Assisted in development, testing, debugging, and maintenance of software applications.
- Worked with senior developers to troubleshoot application issues and improve software quality.
- Gained practical experience with professional software development workflows and the Software Development Life Cycle (SDLC).
- Collaborated with team members on application maintenance, debugging, testing, and technical problem solving.

TECHNICAL SKILLS

Programming AI / ML	Python, C#, .NET, SQL, HTML, JavaScript, Jupyter Notebook Machine Learning, Deep Learning, CNNs, LSTMs, Multimodal Learning, Predictive Modeling, Time-Series Forecasting, Computer Vision, NLP, Fraud Detection
MLOps	FastAPI, Docker, REST APIs, Model Serving, Configuration Management, Logging, Testing, Monitoring, Metrics, Production ML Pipelines
Generative AI	Llama 3, Ollama, LangChain, Local AI Agents, RAG Concepts, LLM-based Automation
Data Engineering	SQL Server, SSIS, ETL Pipelines, Data Ingestion, Web Scraping, CSV Automation, Exploratory Data Analysis
Frameworks / Tools	PyTorch, FastAPI, OpenCV, MediaPipe, Streamlit, Docker, Git, GitHub, VS Code
Developer Tools	CLI Development, GitHub Packages, PyPI Packaging, Automation Scripts, VS Code Extension API

PROJECTS

AQI Forecasting MLOps Platform

10Pearls Internship Project

Python, FastAPI, Streamlit, Docker, MLOps

- Designed and developed an end-to-end machine learning system for **Air Quality Index prediction and forecasting**.
- Implemented modular components for environmental data ingestion, validation, preprocessing, prediction, API serving, and dashboard integration.
- Developed a production-oriented **FastAPI prediction service** and integrated it with a **Streamlit monitoring and visualization dashboard**.
- Implemented robust API-client functionality including asynchronous communication, retries, caching, authentication-ready architecture, and failure handling.
- Incorporated MLOps practices including configuration management, structured logging, automated testing, Docker-based deployment, application metrics, and observability.
- Designed the project using maintainable software engineering principles to support future model updates, monitoring, explainability, and deployment.

DeepGuard – AI-Powered Deepfake Video Detection System

Research-Oriented AI Project

Python, PyTorch, Multimodal AI, Docker

- Developed a research-focused multimodal deepfake detection system for identifying AI-generated and manipulated videos.
- Designed a multi-branch AI architecture combining visual, motion, physics-informed, forensic, and audio-based evidence.
- Worked with model checkpoints, expert branches, multimodal fusion, training pipelines, evaluation metrics, and dataset processing.
- Structured the project with backend, frontend, scripts, saved models, tests, Dockerfiles, and documentation for deployment-oriented development.

Intelligent Data Agent

Local AI Automation Project

Python, Ollama, LangChain, CLI

- Built a local CLI-based AI assistant for automated execution of data-related tasks using local LLM workflows.
- Integrated Ollama and LangChain for privacy-aware and local-first AI automation.
- Focused on secure execution, structured command handling, and developer productivity for data workflows.

InstaPipeline – Automated Data Pipeline

Data Engineering Project

SSIS, SQL Server, ETL

- Designed an automated ETL pipeline for processing Instagram video files and maintaining structured data logs.
- Implemented data extraction, categorization, CSV logging, and automated archive-folder movement.
- Added workflow logic to prevent duplicate file processing and improve pipeline reliability.

SAM-EDA

Open-Source Python Package

Python, PyPI, GitHub Packages

- Developed an open-source Python package for automated heuristic Exploratory Data Analysis.
- Designed the package to reduce repetitive EDA tasks and accelerate early-stage dataset understanding.
- Published and maintained the project through open-source package distribution channels.

Credit Card Fraud Detection

Machine Learning Project

Python, CNN, Kaggle, Streamlit

- Developed a deep learning-based fraud detection system for classifying financial transactions as legitimate or fraudulent.
- Implemented preprocessing, CNN-based classification, model evaluation, and a Streamlit-based demonstration interface.
- Evaluated model performance using accuracy, precision, recall, and F1-score.

Stock Price Prediction – Google

Time-Series Forecasting Project

Python, LSTM, Jupyter Notebook, Streamlit

- Developed an LSTM-based time-series forecasting application for Google stock price prediction.
- Applied historical data preprocessing, sequence generation, deep learning model training, and prediction visualization.
- Developed a Streamlit interface for displaying historical trends and forecasting outputs.

PalmPilot – Gesture-Based Game Controller

Computer Vision Project

Python, OpenCV, MediaPipe

- Built a real-time computer vision application that converts hand gestures into keyboard controls for gaming.
- Used webcam-based hand landmark detection and gesture mapping to control movement and in-game actions.
- Implemented the system using a standard laptop camera without requiring additional hardware.

VS Code Developer Extensions

Developer Productivity Tools

VS Code API, JavaScript, Automation

- Developed custom VS Code extensions for improving developer productivity and workflow automation.
- Built tools including an automated SQL translator and an “Explain Like I Am Five” code explanation extension.

ADDITIONAL PROJECTS

- **PINN Web Application** – Explored Physics-Informed Neural Network concepts and the integration of mathematical constraints with neural networks.
- **Web Data Extraction Tool** – Developed a Python web-scraping utility for extracting, cleaning, transforming, and exporting structured web data to CSV.
- **House Price Prediction** – Built a regression-based machine learning system using preprocessing, feature analysis, model training, and prediction evaluation.
- **Chatbot Application** – Developed a conversational chatbot interface using web technologies and NLP-oriented concepts.
- **Instagram Reel Downloader** – Created a Python utility for automating repetitive Instagram media handling tasks.
- **Personal Portfolio Website** – Designed and deployed a portfolio website showcasing AI, data engineering, machine learning, and software projects.

OPEN SOURCE & GITHUB

- Maintain a public GitHub portfolio containing projects across artificial intelligence, machine learning, computer vision, data engineering, automation, and software development.

- Open-source work includes DeepGuard, PalmPilot, ETL/Data Pipeline, Stock Price Prediction, Credit Card Fraud Detection, House Price Prediction, Web Data Extraction, PINN applications, and developer utilities.
- Experience with Python packaging and distribution using **PyPI and GitHub Packages**.

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

DevOps Bootcamp – 80 Hours

Iqra University

- Gained practical exposure to Git, Docker, deployment concepts, automation workflows, and modern software development practices.

RESEARCH INTERESTS

- AI Safety and Deepfake Detection
- Multimodal Machine Learning
- Computer Vision and Video Forensics
- Production Machine Learning and MLOps
- Local LLM Agents and Retrieval-Augmented Generation
- Data Engineering and Automated ETL Pipelines
- Developer Tools and Workflow Automation

ADDITIONAL INFORMATION

- Strong hands-on project portfolio across artificial intelligence, machine learning, data engineering, computer vision, MLOps, and software development.
- Comfortable working with research-oriented experimentation, applied machine learning notebooks, REST APIs, CLI tools, web applications, data pipelines, and deployment-oriented software architectures.
- Interested in opportunities including **AI/ML Engineer, Machine Learning Engineer, MLOps Engineer, Data Engineer, AI Developer, Software Developer, and Research Assistant**.